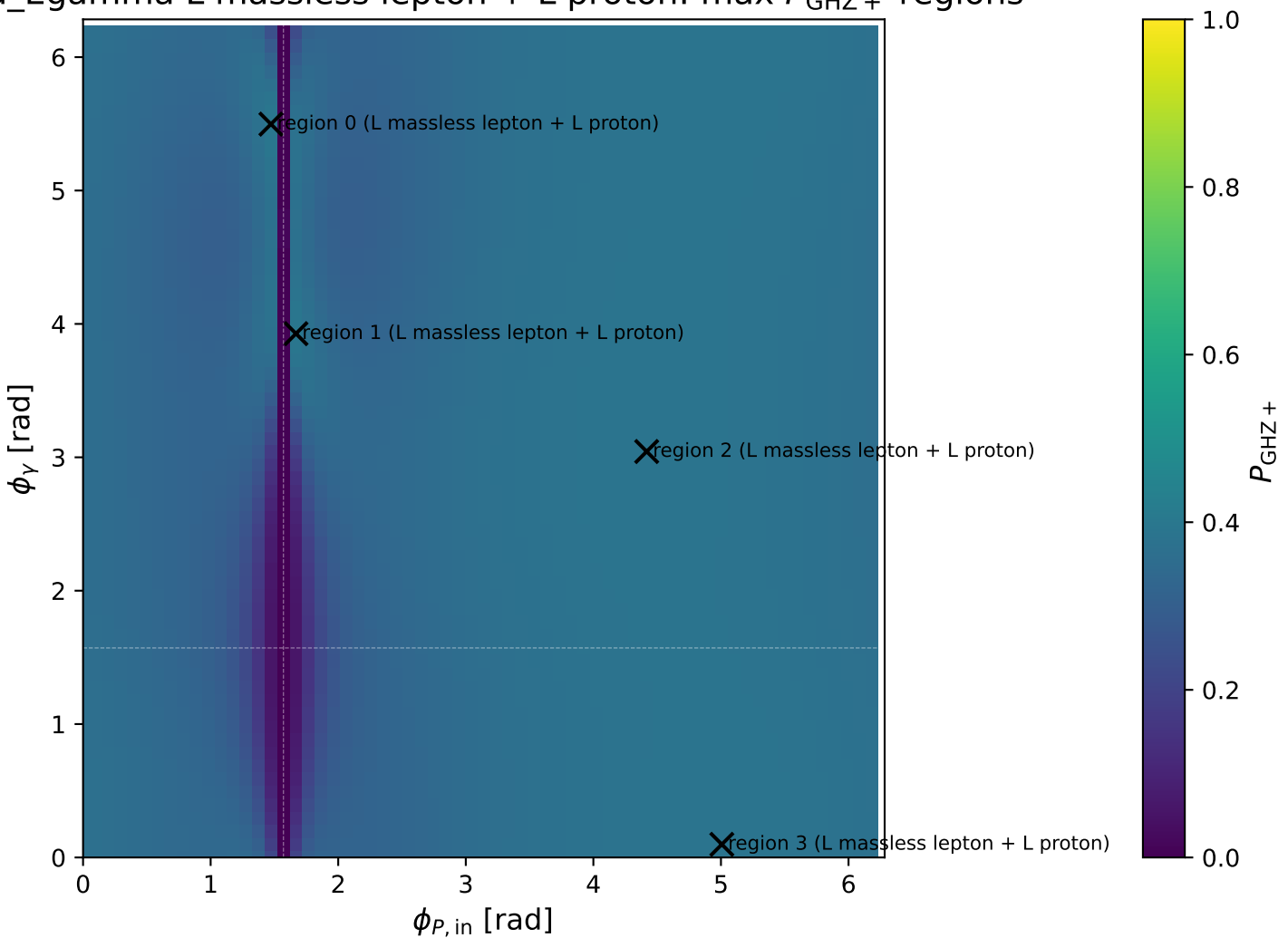
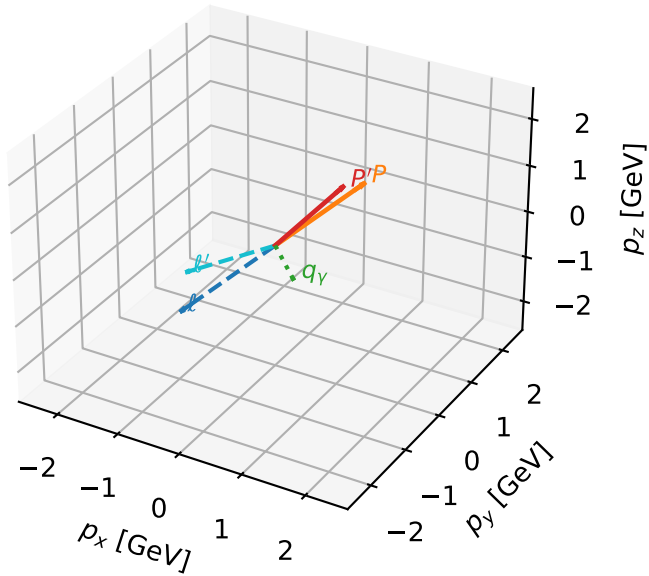


I_{γ} L massless lepton + L proton: max P_{GHZ} + regions



L massless lepton + L proton characteristic max P_{GHZ} + configuration

3D momenta

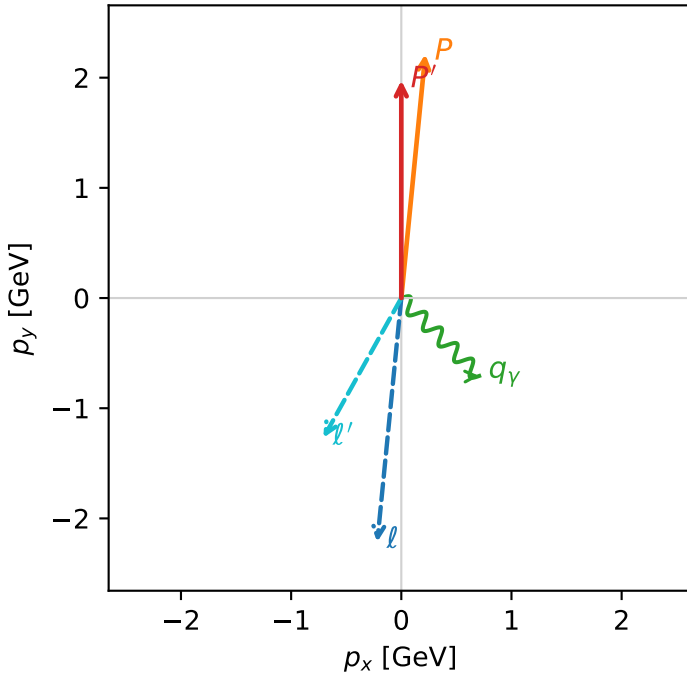


ghz_purity_mid_egamma_ll_region_0 $P_{\{\text{rm GHZ}\}}=0.395388$
 region: mid_Egamma_LL_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

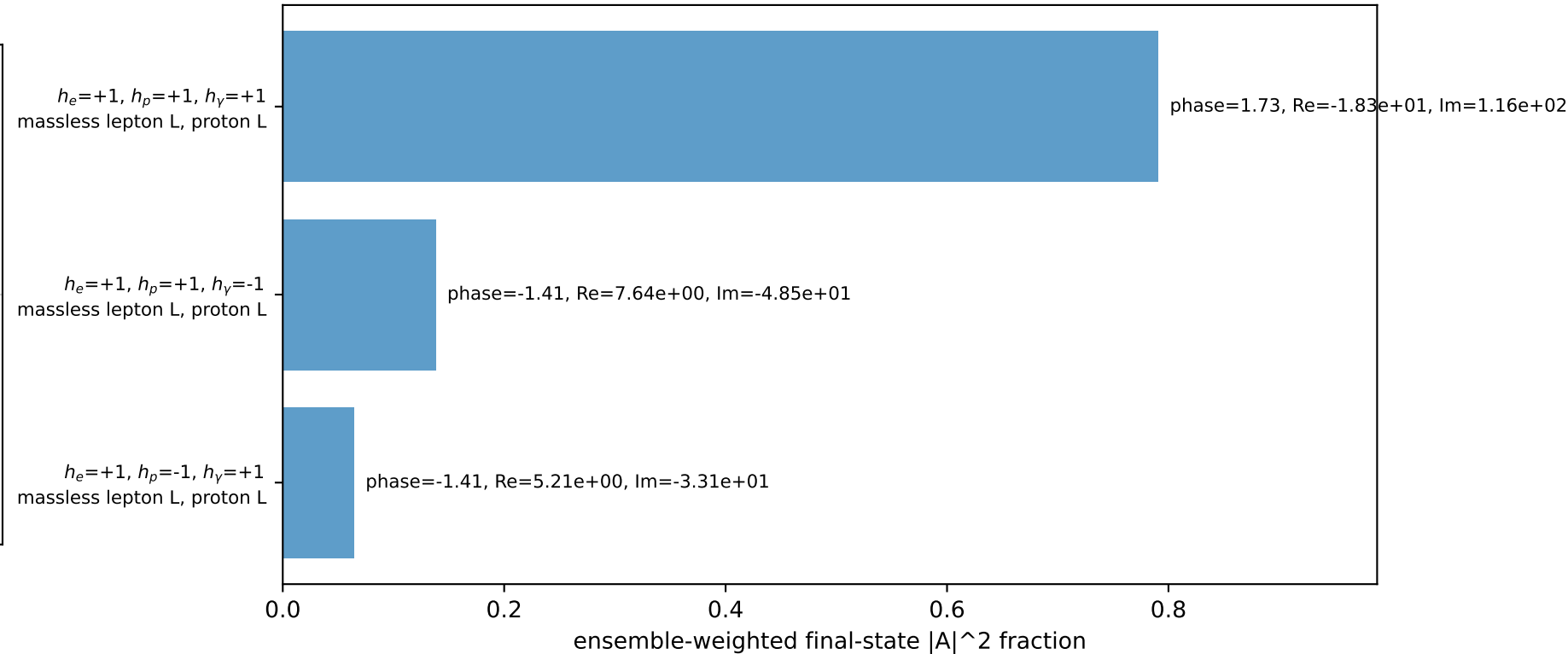
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.9752$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=4.61421$, $\phi_P=1.47262$
 $E_{\gamma}=1$, $\phi_{\gamma}=5.49779$
 $Q^2=0.536623$, $x_B=0.0696625$, $t=-0.0526711$, $W^2=8.0464$, $y=0.373289$
 $\theta(\ell', \gamma)=74.1447$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -0.2180$ $p_y= -2.2137$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.2180$ $p_y= 2.2137$ $p_z= 0.0000$
 ℓ' $E= 1.4519$ $p_x= -0.7071$ $p_y= -1.2681$ $p_z= 0.0000$
 P' $E= 2.1866$ $p_x= 0.0000$ $p_y= 1.9752$ $p_z= 0.0000$
 q_{γ} $E= 1.0000$ $p_x= 0.7071$ $p_y= -0.7071$ $p_z= 0.0000$

Transverse plane

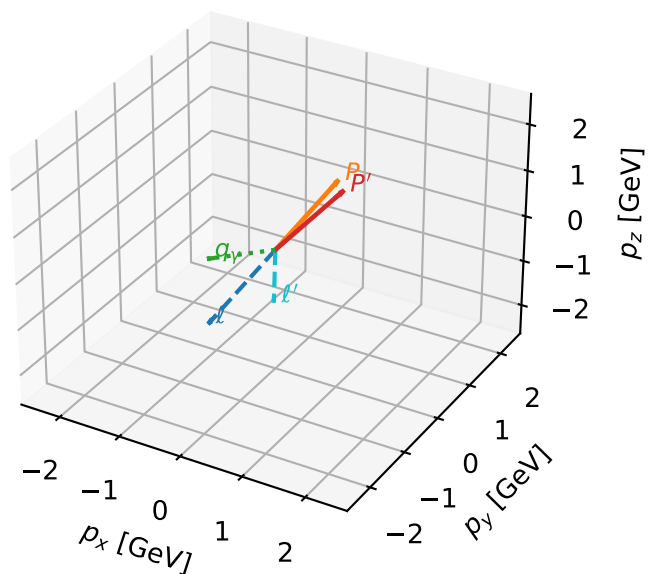


Leading initial-ensemble/final-state components (N=3, retained=99.3%)



L massless lepton + L proton characteristic max P_{GHZ} + configuration

3D momenta

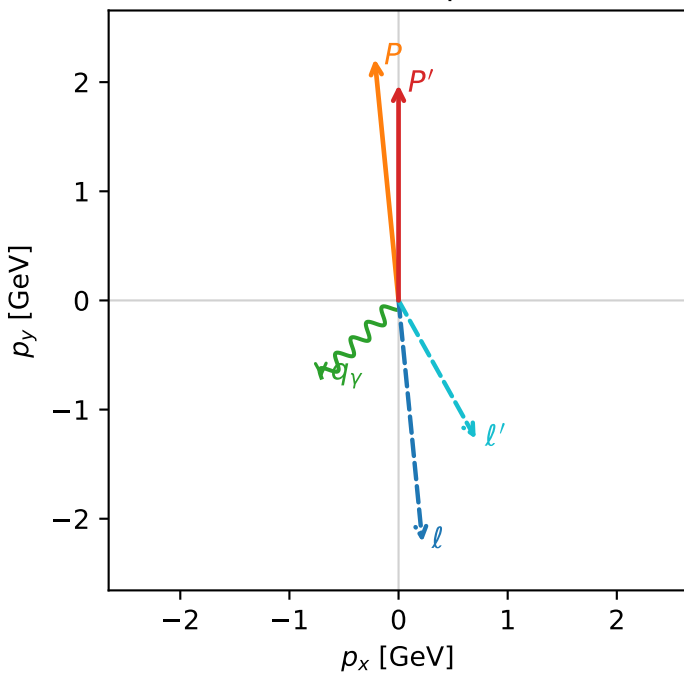


ghz_purity_mid_egamma_ll_region_1 $P_{\{\text{rm GHZ}\}}=0.395388$
 region: mid_Egamma_LL_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

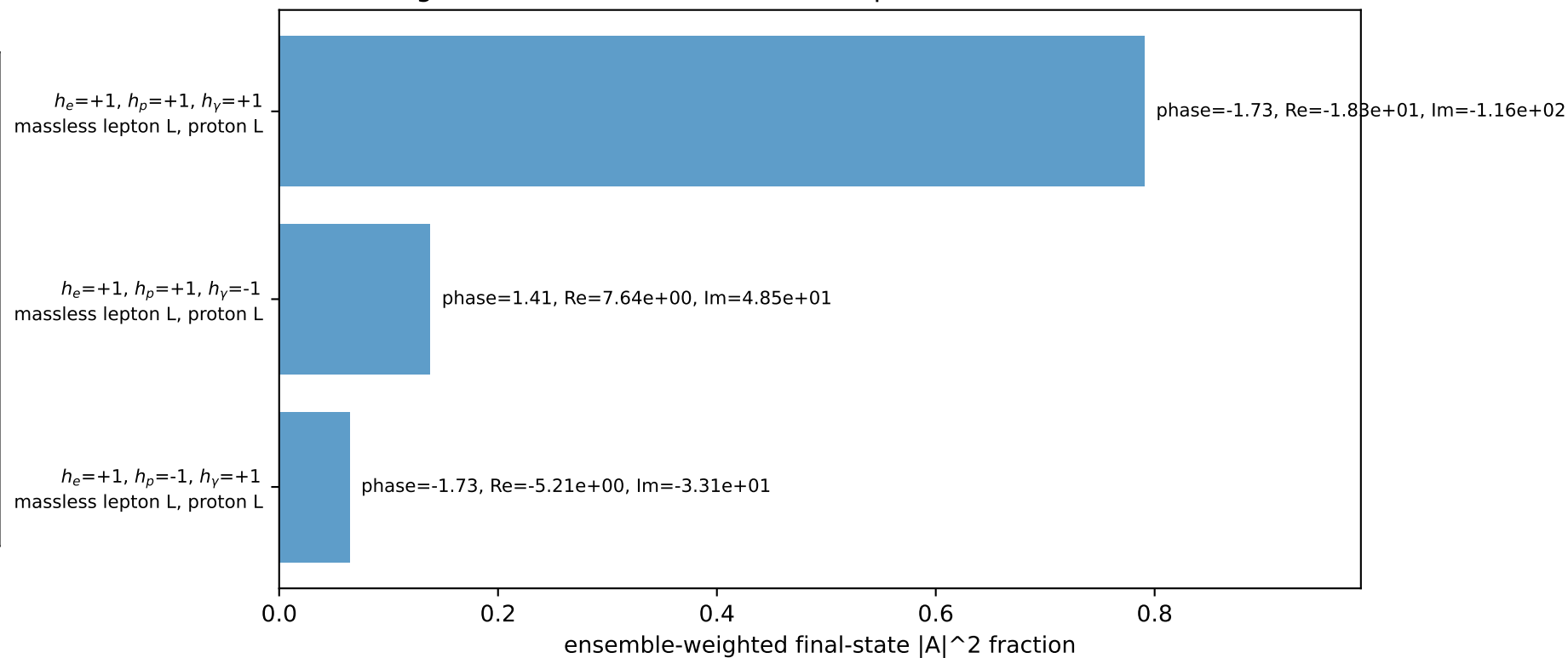
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.9752$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=4.81056$, $\phi_P=1.66897$
 $E_Y=1$, $\phi_Y=3.92699$
 $Q^2=0.536623$, $x_B=0.0696625$, $t=-0.0526711$, $W^2=8.0464$, $y=0.373289$
 $\theta(\ell', \gamma)=74.1447$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 0.2180$ $p_y= -2.2137$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.2180$ $p_y= 2.2137$ $p_z= 0.0000$
 ℓ' $E= 1.4519$ $p_x= 0.7071$ $p_y= -1.2681$ $p_z= 0.0000$
 P' $E= 2.1866$ $p_x= 0.0000$ $p_y= 1.9752$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= -0.7071$ $p_y= -0.7071$ $p_z= 0.0000$

Transverse plane

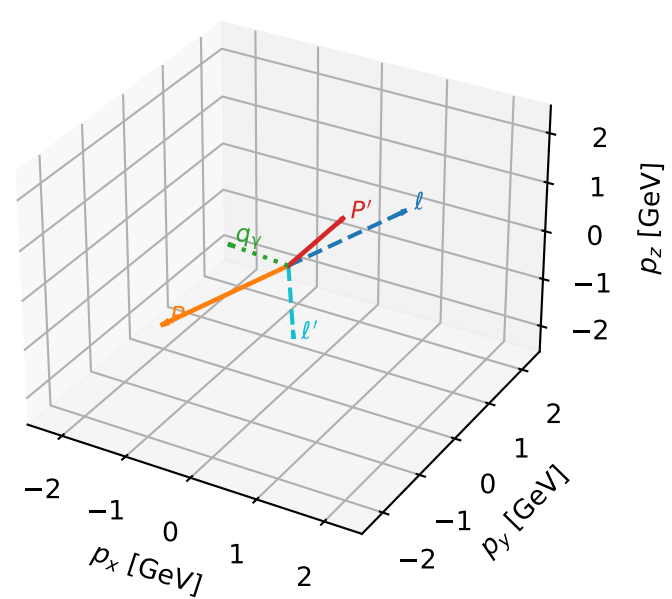


Leading initial-ensemble/final-state components (N=3, retained=99.3%)



L massless lepton + L proton characteristic max P_{GHZ} + configuration

3D momenta

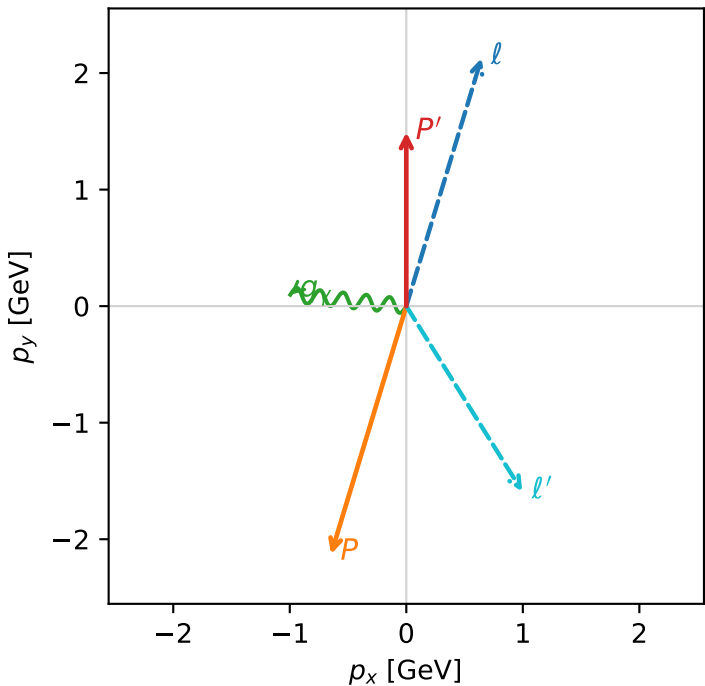


ghz_purity_mid_egamma_ll_region_2 $P_{\text{GHZ}}=0.38561$
region: mid_Egamma_LL_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_{t_0}\rangle \otimes |L_p\rangle$

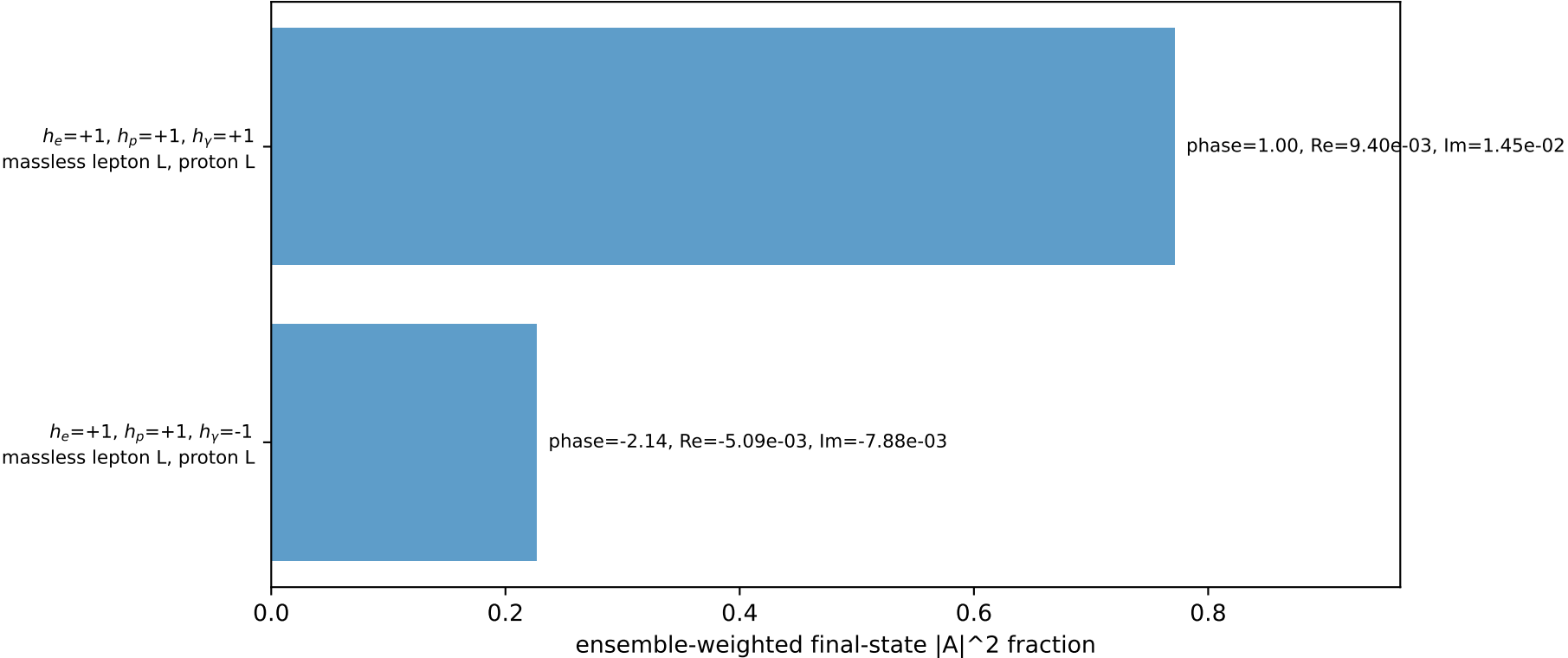
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.49222$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.27627$, $\phi_p=4.41786$
 $E_\gamma=1$, $\phi_\gamma=3.04342$
 $Q^2=13.8308$, $x_B=0.810555$, $t=-13.103$, $W^2=4.11241$, $y=0.826872$
 $\theta(l', \gamma)=127.664$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= 0.6457$ $p_y= 2.1286$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.6457$ $p_y= -2.1286$ $p_z= 0.0000$
 l' $E= 1.8760$ $p_x= 0.9952$ $p_y= -1.5902$ $p_z= 0.0000$
 P' $E= 1.7625$ $p_x= 0.0000$ $p_y= 1.4922$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= -0.9952$ $p_y= 0.0980$ $p_z= 0.0000$

Transverse plane

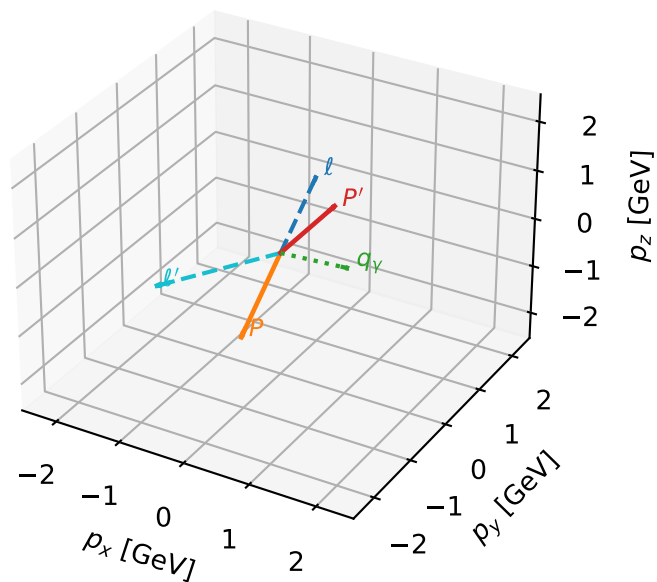


Leading initial-ensemble/final-state components (N=2, retained=99.8%)



L massless lepton + L proton characteristic max P_{GHZ} + configuration

3D momenta



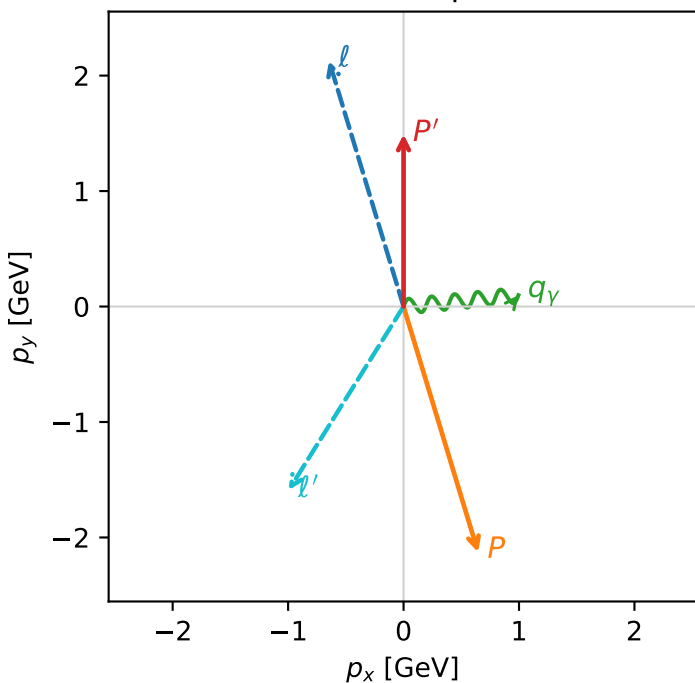
ghz_purity_mid_egamma_ll_region_3 $P_{\text{GHZ}}=0.38561$
 region: mid_Egamma_LL_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.49222$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.86532$, $\phi_P=5.00691$
 $E_\gamma=1$, $\phi_\gamma=0.0981748$
 $Q^2=13.8308$, $x_B=0.810555$, $t=-13.103$, $W^2=4.11241$, $y=0.826872$
 $\theta(\ell', \gamma)=127.664$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E= 2.2244$	$p_x= -0.6457$	$p_y= 2.1286$	$p_z= -0.0000$
P	$E= 2.4141$	$p_x= 0.6457$	$p_y= -2.1286$	$p_z= 0.0000$
ℓ'	$E= 1.8760$	$p_x= -0.9952$	$p_y= -1.5902$	$p_z= 0.0000$
P'	$E= 1.7625$	$p_x= 0.0000$	$p_y= 1.4922$	$p_z= 0.0000$
q_γ	$E= 1.0000$	$p_x= 0.9952$	$p_y= 0.0980$	$p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=2, retained=99.8%)

